INCH-POUND
MIL-R-22307D(SH)
17 September 1992
SUPERSEDING
MIL-R-22307C(SH)
4 September 1974
(See 6.11)

MILITARY SPECIFICATION

RADIAC SET AN/PDR-43( )

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers one type of radiac equipment hereinafter designated as Radiac Set AN/PDR-43( ) which is a lightweight portable equipment for detecting, indicating, and computing gamma radiation in the range between zero and 500 roentgens per hour (R/hr), and for indicating the presence of beta radiation.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, ATTN: SEA 05Q42, Naval Sea Systems Command, 2531 National Center Building 3, Washington, DC 20362-5160 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-E-1	- Electron Tubes, General Specification for.
MIL-E-1/986	- Electron Tube, Radiation Counter Type 7616
MIL-E-1/987	- Electron Tube, Gaseous Discharge Type 7617.
MIL-E-1/988	- Electron Tube, Voltage Regulator Type 7615.
MIL-B-18	- Batteries, Non-Rechargeable, Dry.
MIL-B-18/9	- Battery, Dry, BA-30.
MIL-S-901	- Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for.
MIL-M-10304	- Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification for.
MIL-M-10304/18	- Meters, Multiscale, DC, Panel Type, Ruggedized (Round Flange, 3-1/2 Inch, Style 36).
MIL-E-15090	- Enamel, Equipment, Light-Gray (Formula, No. 111)
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts)· Packaging of.

STANDARDS

MILITARY

MIL-STD-108	- Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-109	- Quality Assurance Terms and Definitions.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-167-1	- Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-415	- Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-781	- Reliability Testing for Engineering Development, Qualification, and Production.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines
MIL-STD-2036	- General Requirements for Electronic Equipment Specifications.

HANDBOOKS

MILITARY

MIL-HDBK-781	- Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production.

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 General. The equipment shall be in accordance with all provisions of MIL-STD-2036, except as otherwise specified herein.

3.2.1 Tropicalization. Tropicalization shall not be required.

3.2.2 Material. Material shall be in accordance with MIL-STD-2036 except as otherwise specified or authorized herein.

3.2.2.1 Magnetic material. The use of magnetic material will be permitted only where a non-magnetic (see 6.5) material cannot be utilized.

3.2.3 Parts. All parts used in the construction of this equipment shall be selected and applied in accordance with MIL-STD-2036. No parts shall be used which require special selection from commercial production runs in order to meet equipment requirements. No mercury or mercury compound shall be used in this equipment. Parts shall be rated at 125 degrees Celsius (°C) or higher if required by circuit application. Plastic encapsulated microcircuits shall not be used.

3.2.4 Shock. The equipment shall withstand the shock tests for grade A, class I, type A, lightweight equipments of MIL-S-901 and shall be energized during the tests. After each blow, the equipment shall continue to operate satisfactorily without internal adjustments and without replacement of any part. There shall be no evidence of collision between any parts of the equipment. Subsequent to the tests, changes in instrument calibration shall not exceed plus or minus 20 percent of initial values.

3.2.5 Vibration. The equipment shall be subjected to the type I vibration test of MIL-STD-167-1 and shall be energized during the test. The instrument shall operate satisfactorily without internal adjustments and without replacement of any part. There shall be no evidence of collision between any parts of the equipment. Any changes in readings during vibration shall not exceed plus or minus 20 percent of the initial value.

3.3 Equipment composition The Radiac Set AN/PDR-43( ) shall include the parts of (a) through (d).

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(a) Radiacmeter.
(b) Shoulder harness for radiacmeter.
(c) 6 polyethylene transparent protective covers for radiacmeter.
(d) Carrying case for radiacmeter

3.3 1 Radiacmeter. The radiacmeter shall be a portable, watertight, battery operated equipment. The equipment shall detect and measure the rate of gamma radiation with a Geiger-Mueller counter intermittently sensitized by the application of pulsed operating voltage. The sensitivity shall be adjustable to correspond to each scale range by appropriate modification of the pulsed operating parameters. The equipment shall compute and integrate the counts and present them in terms of R/hr on an indicating microammeter whose reading shall be proportional to the applied radiation. The equipment shall also detect, but only indicate the presence of beta radiation.

3.3.2 Radiacmeter - functional sections. The radiacmeter shall include a regulated power supply, a Geiger-Mueller tube as radiation detector, a circuit for pulsing the Geiger-Mueller tube, and a computer-indicator circuit. The indicating meter shall also serve for checking the condition of the battery. There shall be a radioactive test sample for checking overall operation of the instrument.

3.3.2.1 Batteries The radiacmeter shall be powered by two type BA-30 batteries in accordance with MIL-B-18 and MIL-B-18/9. Batteries are not required to be furnished with the equipment.

3.3.3 Radiacmeter time constants. The circuit time constants shall be such that for a deflection of 20 percent or more of full scale, the time for the meter to indicate 90 percent of true value shall be between 5 and 10 seconds

3.3.4 Self-noise. No residual or spurious counts shall be introduced by circuitry except from switching.

3.3.5 Indicator. The indicator shall consist of a scale-changing multi-range meter assembly mounted beneath the top cover of the watertight housing. The meter shall be a sensitive-microammeter type MR36M203SPECR in accordance with MIL-M-10304 and MIL-M-10304/18. The meter pointer shall be visible from the top of the radiacmeter through a window The window shall be covered by a transparent material which shall be sealed watertight to the housing. The meter face shall be at the top of the housing, and shall be visible to the operator whether the radiacmeter is carried in the abdominal or hip position by the shoulder harness, or whether carried in the hand. The meter pointer shall not be subject to deviation from electrostatic charges on the meter window.

3.3.5.1 Meter circuit. The meter circuit shall be designed for use with the radiac detector-computer-indicator circuit (see 3.3.1). Adjustable electronic parts shall be included and capable of giving full scale readings in R/hr of 500, 50, 5 on position 3, 4, and 5 as listed herein (see 3 4.1), when using a cobalt 60 (Co^{60}) or cesium 137 (Cs^{137}) gamma ray source in accordance with the calibration method in 3.8.3

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Position or range	1 - OFF 2 - Battery condition 3 - 0-500 4 - 0-50 5 - 0-5

3.3.5.2 Meter illumination. The meter scale shall be illuminated by a concealed lamp within the housing under control of the operator by means of the switch on the top of the case (see 3.6). The brightness of illumination shall be the minimum that will permit reading the scale under conditions of a blackout. The power for illumination shall be supplied by the radiacmeter battery. For determination of overall power requirements, it shall be assumed that illumination will be required intermittently for not more than 10 percent of the time that the radiacmeter is operated.

3.3.5.3 Meter zero adjustment. The meter shall have a zero mechanical adjustment but at a location not accessible to the operator.

3.3.5.4 Meter balance. Meter balance shall be such that the equipment calibration and accuracy specified in 3.8 and 3 9.1 shall apply when the equipment is tilted 45 degrees.

3.3.5.5 Spurious indications. Spurious meter indications due to vibration and shock shall be minimized. The meter indication shall not move more than 10 percent of full scale when the equipment receives light jars similar to a rap against the side with the operator's knuckles and it shall return immediately to its original position.

3.4 Controls.

3.4.1 Range selector switch assembly. A five-position selector switch shall control the necessary circuitry to provide for turning off the radiacmeter; checking the battery condition; selecting the appropriate portion of the operating circuit for specified radiation ranges, and positioning the corresponding meter scales. The switch shall be operated by a pointer-lever located on the top of the housing. Switch positions shall be as specified in 3.3.5.1

3.4.2 Light switch. Control of illumination shall be by a momentarycontact switch on top of the case (see 3.3.5.2).

3.4.3 Function selector assembly. The function selector assembly shall enable performance of the functions in (a) through (c) as follows:

(a) Check of overall operation.
(b) Measurement of gamma radiation intensity.
(c) Detection of beta radiation.

Details of construction shall be subject to the approval of the contracting activity.

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3.4.3.1 Arrangement of function selector assembly. The function selector assembly shall be located inside the radiacmeter housing, except for the control. It shall include a source of beta radiation appropriate for checking overall function (see 3.4.3(a)), together with shields. The parts shall be selectively moveable with respect to one another to provide for the functions of 3.4 3

3.4.3.2 Positions of source and shields. The positions of the source and shields relative to one another shall be as specified in (a) through (c):

(a) CHECK. In the CHECK position, the beta window of the Geiger-Mueller tube shall be shielded from the beta window of the housing but exposed to the check source.
(b) GAMMA. In the GAMMA position, the beta window of the Geiger-Mueller tube shall be shielded from the check source and shielded from the beta window of the housing.
(c) BETA. In the BETA position, the beta window of the Geiger-Mueller tube shall be shielded from the check source but exposed to the beta window of the housing.

3.4.3.3 Control. The control of the function selector shall be by a pointer-lever located on top of the housing The control positions shall be in the following sequence, reading clockwise· CHECK, GAMMA, BETA. The GAMMA position shall be the normal position and the function selector control shall return automatically to and remain in the GAMMA position when released. The three positions of the pointer shall be marked on the housing to indicate the function selected The function selector control shall withstand a torque of 15 inch-pounds on the pointer-lever without slipping

3.4.3.4 Shielding of source. In the GAMMA and BETA positions, the source shall be shielded to prevent the escape of radiation. In the CHECK position, the escape of radiation shall be limited to the direction toward the Geiger-Mueller tube. It shall not be necessary to expose the source in order to obtain access to any part of the radiacmeter except for the function selector assembly.

3.4.4 Beta radiation source. The source shall consist of not more than 100 microcuries of Krypton 85. It shall produce a reading of 0.5 to 3.0 R/hr on the 0 to 5 R scale, as a check of the equipment. The source shall not produce noticeable background reading when the selector is in the GAMMA or BETA position Glass tubulation is not acceptable. Copper or other metal tubulation is required.

3.4.5 Range selector switch positions The order of range selector switch positions of 3 4.5.1 through 3.4.5.5 shall be used.

3.4.5.1 Position 1 - off position. Maximum counterclockwise position. The switch shall open at least the minimum number of circuits that will ensure no current drain from the batteries

3.4.5.2 Position 2 - battery condition. The switch shall connect the meter into the circuit in such a way as to indicate the condition of the batteries under load. The mark BATT located on the meter scale shall indicate the lower limit of battery potential.

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3.4.5.3 Position 3 Position 3 shall be the 0 to 500 R/hr position.

3.4.5.4 Position 4. Position 4 shall be the 0 to 50 R/hr position.

3.4.5.5 Position 5. Position 5 shall be the 0 to 5 R/hr position.

3.4.6 Meter scales. There shall be five scales in the meter, each marked to correspond to one range selector switch position of 3.4.5. Positioning the range selector switch shall cause the corresponding meter scale to be presented to view.

3.5 Electronic chassis.

3.5.1 Subassemblies. The parts constituting the several functional sections shall be grouped into subassemblies according to function and assembled on a lightweight metal chassis as a removable assembly attached to the top cover of the housing. Subject to the approval of the contracting activity, parts and subassemblies shall be arranged to plug into the unit where practical. Construction shall be such that parts and subassemblies can be changed without special tools. Transistors and other semiconductors shall be of the same types to the extent practical. Electron tubes, transistors, and crystal diodes shall be secured so that they can be removed and replaced without disturbing other circuitry.

3.5.1.1 Electron tubes. Electron tubes used in the equipment shall be as follows:

(a) Low sensitivity counter (gamma self-quenching) tube of MIL-E-1 and MIL-E-1/986.
(b) Gaseous. Discharge relaxation oscillator tube of MIL-E-1 and MIL-E-1/987.
(c) Corona Discharge voltage regulator tube of MIL-E-1 and MIL-E-1/988.

3.5.2 Detecting element mounting. The Geiger-Mueller radiation-sensitive detecting tube shall be mounted vertically on the electronic chassis near the front end of the instrument and on the centerline with respect to the right and left sides of the housing. There shall be an opening in the bottom of the housing adjacent to the Geiger-Mueller tube to admit beta radiation. The beta opening shall be covered by beta window permitting detection of beta particles greater than 0.1 Megaelectronvolt (MeV). The beta window shall maintain the watertight integrity of the housing.

3.6 Radiacmeter housing. The housing shall consist of an aluminum enclosure which shall be watertight in accordance with MIL-STD-108. The housing shall be designed for easy and convenient use during survey operations. The shape shall be rectangular, with the smallest dimensions practicable. Form factor shall be subject to approval of the contracting activity. All surfaces shall be smooth The housing shall be of split type construction using a watertight seal and held together by recessed captive screws from the top surface. The housing shall be protected against leaks through the openings for external operating controls, function selector, hold-down screws, calibration access plug, and other openings, by O-ring seals or the equivalent. No drainage plug shall be provided (see 3.5.2 for the beta window).

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3.7 Housing compartmentation. The housing shall be divided into separate watertight compartments for the chassis and for the batteries, respectively. The battery compartment shall be incorporated in the top cover. There shall be separate access to the battery compartment through the top cover to permit changing batteries without opening the chassis compartment. Electrical connections between the compartments shall be watertight insulated feed-through connectors through the battery compartment wall permanently attached to wiring as approved by the contracting activity

3 7 1 Housing materials.

3.7.1.1 Metal. The housing shall be constructed of aluminum as light as possible consistent with the strength required. The protection of aluminum (anodizing) shall be in accordance with MIL-STD-2036.

3.7.1.2 Housing surfaces. All surfaces shall be as smooth as practicable with no sharp edges, crevices, or other surface irregularities to collect contaminated material. Indication markings engraved on the housing shall be filled with nonporous paint of contrasting color so as to produce a smooth surface. Rubber or felt feet shall not be used. Parts subject to contamination shall be capable of decontamination without recourse to laboratory equipment or techniques.

3.7.1.3 Calibration markings. The housing shall be marked with crosslines on the bottom, left, right, and front surfaces to indicate the location of the center position of the Geiger-Mueller detecting element (see 3.5.2) for the purpose of calibration sensitivity The front of the case shall be marked CALIBRATION CENTER. Markings for calibration shall be as approved by the contracting activity

3 7.1 4 Handcarrying handle The radiacmeter shall be equipped with a retractable handle for carrying the instrument in one hand. The balance point of the instrument shall be located at the center of the handle. The handle shall be retractable to lie flat on the top of the housing but only when the operating switch is at OFF position. It shall be extendable so as to be locked at the right of, left of, or above the housing by the operator. The requirements as to smooth surfaces of 3.7.1.2 apply also to the handle and the locking mechanism.

3.8 Calibration.

3.8.1 Calibration controls. The minimum number of controls shall be furnished to provide for complete and independent adjustment of all three scales in accordance with the requirements of 3.4.5.3, 3.4 5.4, and 3.4.5.5 All calibration controls shall be accessible through ports on top of the housing. The calibration controls shall be of such value that they are set at midrange when the instrument is calibrated (see 3.8.4) and provide for at least a plus or minus 20 percent variation on either side of the correct calibration value.

3.8.2 Calibration standards. Cobalt 60 or cesium 137 shall be the standards of reference for gamma calibration. The calibration sources used shall be approved by the contracting activity. The gamma intensities shall be traceable to the National Institute of Standards and Technology.

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3.8.3 Calibration method. For the purpose of calibration of the equipment, the radiation shall be directed normal to the surface marked CALIBRATION CENTER. Distances shall be measured to the center position of the Geiger-Mueller detector (see 3.7.1.3).

3.8.4 Calibration settings. The range settings and radiation field intensities for calibration shall be as specified in table I.

TABLE I. Calibration settings.

Selector switch	Desired meter reading	Co^{60} field intensity	Cs^{137} field intensity
500 50 R/hr 5	400 40 R/hr ±3 percent 4	400 40 R/hr ±3 percent 4	472 47.2 R/hr ±3 percent 4 72

Two additional readings shall be made on each scale, at approximately 40 percent and 15 percent of full scale value.

3.8.5 Directionality. The readings shall be within plus or minus 20 percent of the true value of radiation intensity for orientations of the radiacmeter within 45 degrees as measured from the center position of Geiger-Mueller tube and the line normal to the calibration center (see 3.8.3).

3.9 Operating characteristics.

3.9.1 Accuracy. Any combination of errors introduced by the factors of table II shall not cause any reading of this equipment to deviate more than plus or minus 20 percent from the true value of radiation intensity

TABLE II. Operational characteristics for accuracy.

Tests	Applicable paragraph
Battery life	3.9 3
Temperature	3.9.4
Humidity	3.9.5
Altitude	3 9 6
Shelf life	3.10
Calibration range	3.8.4

3.9.2 Accuracy restriction The accuracy requirement shall apply between 10 and 100 percent of the full scale ranges specified in 3.3.5.1

3.9.3 Battery life The required accuracy shall be maintained for at least 40 hours when the radiacmeter is operated continuously for a period of at least 40 hours by one set of batteries, with the indicating meter lamp on at least 10 percent of the time.

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3.9.4 Temperature. The temperature classification for the radiac set shall be range 3 in accordance with MIL-STD-2036 An external power source may be used in temperature tests.

3.9.5 Humidity. The equipment shall operate satisfactorily within the accuracy limits specified in 3.9.1 when subjected to the humidity tests specified in MIL-STD-2036. The upper temperature limit for the humidity test shall be plus 50°C.

3.9.6 Altitude. The equipment shall perform satisfactorily at pressures equivalent to any altitude up to and including 50,000 feet (15,240 meters).

3 10 Shelf life. The instrument shall be operative and shall maintain accuracy without recalibration following a storage period of 30 days.

3 11 Saturation effects. While the radiacmeter is subjected to dose-rates in the range 600 to 10,000 R/hr, it shall indicate off-scale readings at the high end of each scale for position 3 (see 3 4.5.3), position 4 (see 3.4.5 4), and position 5 (see 3.4 5 5).

3 12 Energy dependence No reading of this equipment between 10 and 100 percent of the full scale deflection of the scale ranges specified in 3 4 6 shall deviate more than plus or minus 20 percent from the true value of radiation intensity for any one energy over the gamma energy range of 0.08 to 1.3 electron volts MeV. Radiation for energy dependence determinations shall be directed normal to the calibration center marking (see 3.7.1.3) or in any direction within 45 degrees as measured from the center position of the Geiger-Mueller tube and the line normal to the calibration center. Subject to the approval of the contracting activity, energy response of the equipment at 50 kilo-electronvolts (KeV) shall be 30 to 60 percent of the equipment response at 80 KeV The manner of energy dependence shielding shall be subject to the approval of the contracting activity. The shield shall not interfere with access to the Geiger-Mueller tube.

3.13 Beta response. Beta indication above 0.1 MeV shall be provided but the equipment will not necessarily be calibrated for beta radiation.

3 14 Weight and volume. The radiacmeter, less carrying case, but complete with batteries and shoulder harness, shall weigh not more than 5 pounds (2 268 kilograms) and the volume shall not exceed 150 cubic inches (2,458 cubic centimeters).

3.15 Shoulder harness. A shoulder harness shall be provided for carrying the radiacmeter on the abdomen or hip. The material employed shall be nonporous and shall resist absorption of water or other material bearing radioactive contamination. The harness shall be attached to the radiacmeter by removable connectors, which shall not be subject to accidental detachment during operational use. Design of the shoulder harness shall be subject to approval of the contracting activity.

3.16 Plastic cover. The plastic cover shall enclose the radiacmeter for protection against contamination by dust or liquids and shall permit effective manipulation of the controls and reading of the meter during use

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The material of the cover shall be transparent, shall produce no appreciable change in gamma calibration, and shall be usable between minus 20 and plus 50°C. It shall be flameproof and shall not support combustion.

3.17 Carrying case.

3.17.1 Description. The carrying case shall be a lightweight case conforming to MIL-STD-108 for splashproof enclosures for other than motors and generators. One folding handle of a type not to interfere with stacking shall be provided for convenience in carrying. The surfaces shall be as smooth as practicable, with the minimum amount of crevices to collect contamination. The material shall be drawn aluminum alloy in accordance with MIL-STD-2036.

3.17.2 Function. The carrying case shall protect the radiacmeter from damage due to shock and vibration encountered in handling. It shall house the radiacmeter, one set of spare batteries, the shoulder harness, two technical manuals, and six transparent plastic covers.

3.17.3 Weight and volume. The carrying case, less the radiacmeter, shall weigh not more than 3.5 pounds and the volume shall not exceed 300 cubic inches (4,916 cubic centimeters).

3.17.4 Instruction plate. An instruction plate shall be provided inside the carrying case to list all the items which belong in the carrying case, with quantity and location. For spare batteries, however, only the stowage location shall be shown. The plate shall be visible when the carrying case cover is open. The instruction plate shall conform to the identification and marking requirements of MIL-STD-2036.

3.18 Finish Protection of aluminum surfaces (anodizing) shall be in accordance with MIL-STD-2036. Paint for the housing shall be enamel, class 1 of MIL-E-15090. The finish shall not chip from a sharp hammer tap

3.19 Reliability. The radiac set shall have a minimum mean-time-between-failure (θo as defined by MIL-STD-781) of 1000 hours at any combination of specified operating conditions.

3.20 Marking for radioactivity. Marking for radioactivity shall be in accordance with MIL-STD-129. The markings on the radiacmeter housing and on the carrying case shall be on the same side as the identification plate.

3.21 Maintenance. Test points and test features are required in accordance with MIL-STD-2036 and MIL-STD-415.

3.22 Qualitative maintainability requirements. The equipment shall be constructed to provide ease of maintenance, accessibility, and replacement of replaceable assemblies and parts. Ease of maintenance, troubleshooting and repair shall be a primary design consideration.

3.23 Electromagnetic compatibility. The radiac set shall meet the requirements of MIL-STD-461, tests RE01, RE02, RS02, and RS03. The limits for test RS03, however, shall be as specified in table III.

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TABLE III. Frequency and field intensity for electromagnetic compatibility.

Frequency (megahertz (MHz))	Field intensity (volts per (V/ru))
0.014 - 0.30	1 0 Average
1/ 0.30 - 1.5	300 Average
1/ 1.5 - 30	100 Average
30 - 12,400	10 Average

1/ Test at only three discrete frequencies within these ranges.

3 24 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified

3.25 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.26 Workmanship. Workmanship shall be in accordance with MIL-STD-2036.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5 The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program (see 6.3). The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Quality assurance terms and definitions Quality assurance terms used in this specification shall be as defined in MIL-STD-109 (see 6 3)

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. Unless otherwise specified, three units shall be required for first article inspection. First article inspection shall consist of the examinations and tests specified in 4.3.1 through 4.3.1.2 (see 6.3).

4.3.1 First article inspection shall be as specified in MIL-STD-2036, except that the tests of (a) through (f) shall not apply.

(a) Supply line voltage and frequency.
(b) Power.
(c) Water cooling.
(d) Accelerated life.
(e) Inclination and noise.
(f) Reliability.

4 3.1.1 The tests of (a) through (g) shall be added.

(a) Battery power (see 3.9.3).
(b) Accuracy (see 3.9.1 and 3.9.2).
(c) Energy dependence (see 3.12).
(d) Saturation (see 3.11).
(e) Circuit response time (see 3.3.3)
(f) Low pressure (altitude) (method 500, procedure I of MIL-STD-810).
(g) Calibration control test (see 3.8.1 and 3.8.4).

4.3.1.2 Electromagnetic compatibility test The requirements of 3.23 shall be verified by tests performed in accordance with MIL-STD-462. Tests shall be performed on one unit only.

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4.4 Quality conformance inspection. Quality conformance inspection shall consist of production inspection (see 4.4.1) and production control inspections (see 4.4.2).

4.4 1 Production inspection. Production inspection shall be in accordance with MIL-STD-2036 and, in addition, shall include calibration for gamma at three points on each scale in accordance with 3.8.4 and check for beta on each scale (see 3.13).

4.4.1.1 Break-in period. Prior to production inspection, each equipment shall be operated for 72 hours in the counting mode, 24 hours on each scale range. For this purpose, the test sample may serve as the source and an external power supply may be used

4.4 2 Production control inspection Production control inspection including sampling, shall include the inspection items in table IV Production control inspection shall be performed on equipments that have passed production inspection. The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment Sampling for production control inspection shall be in accordance with table V.

TABLE IV. Production control inspection

Inspection items	Requirement
Weight and volume	3.14, 3.17.3
Temperature test	3.9.4
Enclosure	3.6, 3.17.1
Controls and control circuits	3.4, 3.3.5
Magnetic test	3 2.2.1
Accuracy	3.9 1
Battery life	3.9.3
Energy dependence	3 12
Electromagnetic compatibility	3.23

TABLE V Sampling for production control inspection.

Lot size	Sample size	Accept	Reject 1/2/3/
2 - 25	3	0	1
26 - 50	5	0	1
51 - 90	6	0	1
91 - 150	7	0	1
151 - 280	10	0	1
281 - 500	11	0	1
501 - 1200	15	0	1
1201 - 3200	18	0	1
3201 - 10000	22	0	1
10001 - OVER	29	0	1

1/ All defective items must be replaced with acceptable items prior to lot acceptance.
2/ Inspect sample size until reject criteria is reached
3/ Rejected lots may be screened and resubmitted for inspection and retest.

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4.4.2.1 Rejected lots. If an inspection lot is rejected, the contractor may also rework a rejected lot to correct the defective units. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.2.2 Reinspection of conforming production control test units. Unless otherwise specified, sample units which have been subjected to, and passed, production control inspections may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all damage.

4.5 Reliability tests. Reliability demonstration shall be performed in accordance with MIL-HDBK-781 and MIL-STD-781.

4.5.1 Reliability demonstration. Reliability demonstration shall be performed on at least four production units. Temperature cycling shall be between minus 40 and plus 50°C. The vibration shall be 1G ± 10 percent peak acceleration value at any nonresonant frequency between 20 and 60 hertz (Hz) measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time. When the equipment being tested contains circuit boards or cards, vibration shall be normal to the plane of the majority of the cards Otherwise, the direction of vibration is not critical The vibration requirement may be met by mounting the equipment solidly (that is, without the equipment shock vibration mounts) to a strong flat plate which is supported by vibration isolators The desired amplitude and frequency may be obtained by a simple vibration machine with an asymmetric weight on a shaft

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attached to the plate and driven by a suitable motor. It is recommended that the vibration equipment be checked for proper operation each 24 hours of operation, and that the vibration transducer be on and monitored continuously during vibration. The voltage shall be the normal specified voltage plus 5, minus 2 percent. The equipment on and off cycle shall be such that the equipment is on during the heating cycle and off during the cooling cycle. Testing shall proceed to a decision in accordance with the criteria specified for test plan V-D of MIL-HDBK-781 (see 6.3).

4.5.2 Reliability sampling tests. Reliability sampling tests shall be conducted throughout production to ensure continued manufacture of acceptable equipment. The sample size shall be in accordance with MIL-HDBK-781 for each 3 months' production. Temperature shall be between 50 and 55°C. Vibration, input voltage and equipment cycling shall be the same as in 4.5.1. Testing shall be in accordance with test plan VI-D of MIL-HDBK-781 (see 6.3).

4.5.3 Preventive maintenance Battery replacement may be performed on reliability test samples at the minimum time interval specified in 3.9.3. No additional preventive maintenance shall be allowed during reliability testing. At the contractor's option, an external power supply may be used in lieu of batteries during the sampling phase of reliability testing

4.5.4 Equipment burn-in. The contractor may elect to operate the reliability test samples for a certain time period prior to the initiation of reliability testing. Any such burn-in applied to test samples shall be applied to all items of equipment submitted for acceptance. Failures occurring during this period may be considered non-relevant, but shall be recorded Burn-in time for all items of equipment submitted for acceptance shall be at the same levels of environment and for the same integral number of cycles as the burn-in conducted on the test samples

4.5.5 Failure. Failure shall be defined as any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements set forth in this specification All failures shall be considered relevant unless determined by the contracting activity to be caused by a condition external to the equipment under test and not encountered in service

4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.9.)

5.1 Preservation, packing, and marking. Unless otherwise specified (see 6.2), packaging shall be in accordance with the applicable levels of preservation and packing specified in MIL-E-17555.

5.2 Unitized loads. When applicable, items packed as specified shall be palletized or containerized in accordance with MIL-STD-147 whenever quantities to one destination equal 40 cubic feet or more. Quantities of packed items too small or impracticable for a unitized load shall be consolidated in shipping containers as specified in 5.1. Palletized loads shall be strapped to the pallet. Unitized loads shall not be subjected to rough handling tests. When applicable, items for shipboard end destinations shall be unitized by hull number

5.3 Marking. In addition to any special marking required (see 6.2), each unit package and exterior container shall be marked in accordance with MIL-STD-129. Marking shall include special marking for method II packs.

5.4 General. Exterior containers shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent possible Unless otherwise specified (see 6.2), loose-fill polystyrene shall not be used for preservation or packing. Adhesives shall not be used in the closure of wraps, bags, or boxes.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Radiac Set AN/PDR-43( ) is intended for use on shipboard and ashore to survey radiological hazards.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
(c) First article inspection, if required (see 3.1).
(d) Number of first article samples to be submitted if other than specified (see 4.3).
(e) Levels of preservation, packing, and marking required (see 5.1 and 5.3)
(f) When loose-fill polystyrene may be used (see 5.4)
(g) When technical manuals are required (see 6.4).

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6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.1.1	DI-MISC-80678	Certification/data report	----
4 1.2	DI-R-5297	Quality program plan (QPP)	----
4.3	DI-NDTI-80603	Test procedures	First article
4 3	DI-NDTI-80809	Test/inspection reports	First article
4.5.1	DI-R-7082	Reliability predictions report	----
4.5.2	DI-RELI-80250	Reliability test plan	----
4.5.2	DI-RELI-80252	Reliability test reports	----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 First article When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation

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6.5.1 Tests, test equipment, and test procedures. Test procedures should include test equipment and tolerance limits to be used for first article inspection and quality conformance inspection. Written test procedures should be as specified in 6 3 herein.

6.5.2 Non-standard parts. The first article test report should include a list of non-standard parts approved or pending approval for the equipment

6.6 Nonmagnetic material. Nonmagnetic material is defined as material which has a magnetic permeability of less than 2.0 after fabrication

6.7 Acceptance levels. The suggested lot acceptance and rejection criteria for production control inspection (see 4.4.2) are given in table V.

6.8 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract

6.8.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified

6.9 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.10 Subject term (key word) listing.

Electron tube
Geiger-Mueller tube
Pulse generator
Radioactive material

6 11 Changes from previous issue Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6665-N567)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given

2 The submitter of this form must complete blocks 4, 5, 6, and 7

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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER	2 DOCUMENT DATE (YYMMDD)
	MIL-R-22307D(SH)	1992 09 17

3. DOCUMENT TITLE

Radiac Set AN/PDR-43 ( )

4 NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)*

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a NAME TECHNICAL POINT OF CONTACT (TPOC): Mr. R. Rhodes (SEA 04RL)	b TELEPHONE (Include Area Code) (1) Commercial: (703)602-2753 (2) AUTOVON: 8-332-2753
c ADDRESS (Include Zip Code) Commander, SEA 05Q42 Naval Sea Systems Command 2531 National Center Building 3 Washington, DC 20362-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 Previous editions are obsolete 198-290

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